Introduction to Software Engineering

Project Proposal & Plan

The student team is required to complete the **Project Proposal & Plan** documentation for the assigned course project, following the attached template.

Software Engineering Department
Faculty of Information and Technology
University of Science

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Project Proposal & Plan

Objectives

This document focus on the following topics:

- ✔ Completing the Project Proposal & Plan document with the following sections:
  - Vision
  - Proposed Solution
  - Project Plan
  - Risk Management
  - Tools & Collaboration Setup
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Project Proposal & Plan document.

1 Member Contribution Assessment

[List every member. Contribution (%) must add up to 100%. Evidence: links to tasks on the board, commits or pull requests.]

ID	Name	Role(s)	Contribution (%)	Evidence
23127520	Nguyễn Phan Thế Vinh	Team lead / PM, Tester	25%	Tasks V1–V3 on task board; PA1 project plan, phase gates, WBS & QA strategy
24127469	Nguyễn Thanh Thảo Nguyên	Business analyst	25%	Tasks N1–N2 on task board; Vision, problem statement, scope definition & stakeholder needs
24127324	Vũ Thị Ánh	Designer / Architect	25%	Tasks A1–A2 on task board; Preliminary architecture, tech stack evaluation & Gantt schedule
24127228	Trần Hà Anh Quân	Developer	25%	Tasks Q1–Q2 on task board; Repository setup, GitHub Projects board & tooling configuration

2 Vision

2.1 Preliminary Problem Statement

2.1.1 Current State and Background

Traditional library management in small-to-medium university libraries or departmental resource centers often relies on fragmented manual procedures, paper logbooks, or disconnected desktop spreadsheets. Managing library operations through these legacy methods leads to several key challenges:

Catalog and Physical Copy Discrepancies: Librarians struggle to maintain accurate synchronization between book metadata and actual physical inventories. Tracking whether a specific copy is available, borrowed, or misplaced requires manual shelf audits and time-consuming record cross-checking.

Inefficient Circulation and Manual Tracking: Over-the-counter checkouts and returns managed via manual logs lack automated due-date calculations and proactive overdue flags. This often results in uncollected late fees, misplaced books, and unrecorded loans.

Friction in Discovery and Holds: Students and academic staff face difficulties searching for resources across multiple criteria (such as ISBN, author, or category) and lack real-time visibility into book availability. When all physical copies of a requested title are checked out, there is no centralized mechanism to place requests or hold queues.

Administrative and Role Management Bottlenecks: Managing user access levels, permissions, and borrower profiles manually creates security risks and operational delays during peak academic periods.

2.1.2 Proposed System: UniLib

UniLib is designed as a lightweight, centralized, web-based University Library Management System that addresses these operational friction points. The system unifies catalog search, physical inventory management, circulation, and reservations into a responsive web interface tailored for Students/Readers, Librarians, and System Administrators.

The scope of UniLib focuses strictly on six core must-have features (F01–F06):

- F01 Authentication & Roles: Secure access control for Readers, Librarians, and Administrators.
- F02 Book Catalog Management: Centralized management of book titles, metadata, and classification.

- F03 Book Copy Management: Tracking physical item instances and real-time status (Available/Borrowed).
- F04 Search & Browse: Multi-attribute filtering (title, author, category, ISBN) for easy discovery.
- F05 Borrow & Return: Circulation tracking with automated due-date assignment and overdue detection.
- F06 Book Reservation: First-In, First-Out (FIFO) queuing mechanism for unavailable titles.

2.1.3 Operating Environment

Client Side: Modern, standard-compliant HTML5 web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) operating on desktop and laptop devices.

Server & Cloud Side: Web application backend and database hosted on cloud container/PaaS platforms (Vercel for frontend hosting, Render for Node.js REST API and PostgreSQL database instances).

2.1.4 Design & Implementation Constraints

Programming Language & Core Tech Stack:

- Frontend: React (JavaScript/TypeScript) rendered in modern browsers.
- Backend: Node.js with Express framework delivering RESTful API endpoints.
- Database: PostgreSQL relational database with structured tables adhering to 3NF.

1.Supported Web Browsers:
Full compatibility required for evergreen HTML5-compliant desktop browsers (Chrome, Firefox, Edge, Safari). Mobile native apps and responsive layouts tailored specifically for small mobile screens are outside the current scope.

2. Documentation Standards:
All specifications, meeting notes, and architecture docs must be maintained in academic-standard Markdown within the official GitHub repository (/docs) and rendered via standard PDF/HTML exports.

3.Scope Boundaries:
Strictly limited to physical book management and core circulation (F01–F06). Out of scope: digital e-books/DRM, online payment integration, hardware integrations (barcode/RFID scanners), inter-library loans, AI recommendation systems, production SMS/email notifications, and distributed multi-campus deployments.

2.1.5 Domain Glossary

Term	Definition

Book Title	The abstract literary work entity containing general metadata such as title, author, publisher, publication year, category, and ISBN.
Book Copy	The physical copy/instance of a specific Book Title present in the library inventory, tracked via a unique copy identifier and an individual operational status (Available, Borrowed, Maintenance).
Loan	The official circulation record documenting the checkout transaction of a specific Book Copy to an authorized Reader, containing issue date, expected due date, and return timestamp.
Reservation	A queue-based request placed by a Reader to hold a Book Title when all associated Book Copies are currently borrowed, processed on a First-In, First-Out (FIFO) order upon copy return.
Overdue	The state of an active Loan transaction that has exceeded its designated return due date without the Book Copy being returned to the library.

2.1.6 Team Assumptions & Items for Clarification

- **Assumptions**

Loan Duration: Default loan period is assumed to be 14 calendar days for student borrowers unless configured otherwise by an Administrator in F01/F05.
Account Provisioning: Reader profiles are either pre-registered via batch university import or manually created by Librarians/Administrators during onboarding.

- **Items for Team Confirmation**

[ ] Loan Limits: Should we enforce a maximum threshold of active loans per reader (e.g., maximum 5 books borrowed simultaneously)? (Need confirmation from Vinh / PM)
[ ] Reservation Expiry: How many days does a reader have to pick up a reserved book once a copy becomes available before it passes to the next person in line? (Need confirmation from Vinh / PM)

2.2 Product Position Statement

Element	Content
For	Small to medium-sized university libraries, academic department libraries, students, and library staff
Who	Need a streamlined, web-based system to manage catalog metadata, track physical book copy availability, automate circulation operations, and handle title hold reservations without complex administrative overhead
The (product name)	Web-based University Library Management System
That	Provides an intuitive, automated platform for multi-attribute book search, real-time physical copy status tracking, seamless borrow/return workflows, and FIFO reservation queuing
Unlike	Manual spreadsheet/paper-based tracking systems, or overly complex, heavy enterprise ILS platforms like Koha or FOLIO that require high maintenance and setup effort
Our product	Focuses on essential core circulation workflows with a modern, responsive user interface, zero-configuration web access, and strict data consistency between book titles and physical copy states

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	Course Lecturer & Teaching Assistants *(Project Sponsor)*	Academic supervisors overseeing the Introduction to Software Engineering course and evaluating project deliverables.	Strict adherence to PA0–PA5 scope, quality of software engineering documentation, clean web architecture, automated testing, and a smooth live demonstration.
2	Reader / Student *(End User)*	University students and faculty members who search for academic books, view availability, and borrow/reserve titles	Fast multi-attribute catalog search, transparent physical copy availability, easy book reservation placement, and a personal dashboard for active loans and due dates.
3	Librarian *(End User)*	Library staff managing catalog records, maintaining physical book copy inventories, and processing circulation Desk tasks	Efficient CRUD operations for book titles and copies, fast checkout/check-in workflows, clear copy state transitions (Available/Borrowed), and overdue tracking
4	Administrator *(End User)*	System administrators responsible for system access setup, role management, and overall platform security	Secure user authentication, role-based access control (RBAC), user account lifecycle management, and system data integrity
5	Team T05 Development Team *(Internal Stakeholders)*	The 4-member student engineering team (PM/Tester, BA, Designer/Architect, Developer) building the UniLib system	Feasible scope execution within 6–8 hours/week/member, high code quality, automated test coverage, and successful project submission

2.4 Scope and Key Features

[List the main features with a MoSCoW priority. Only Must-have features are required to be fully implemented in PA5. State clearly what is out of scope.]

ID	Feature	Description	Priority (M/S/C/W)
F01	User Authentication & Roles	Secure login/logout system with role-based access control (RBAC) supporting Reader, Librarian, and Administrator actors	Must
F02	Book Catalog Management	Comprehensive CRUD capabilities for Librarians to manage abstract book metadata, including ISBN, title, author, category, publisher, and publication year.	Must
F03	Book Copy Management	Tracking of individual physical book items with unique copy barcode/IDs and managing state transitions between Available, Borrowed, and Reserved.	Must

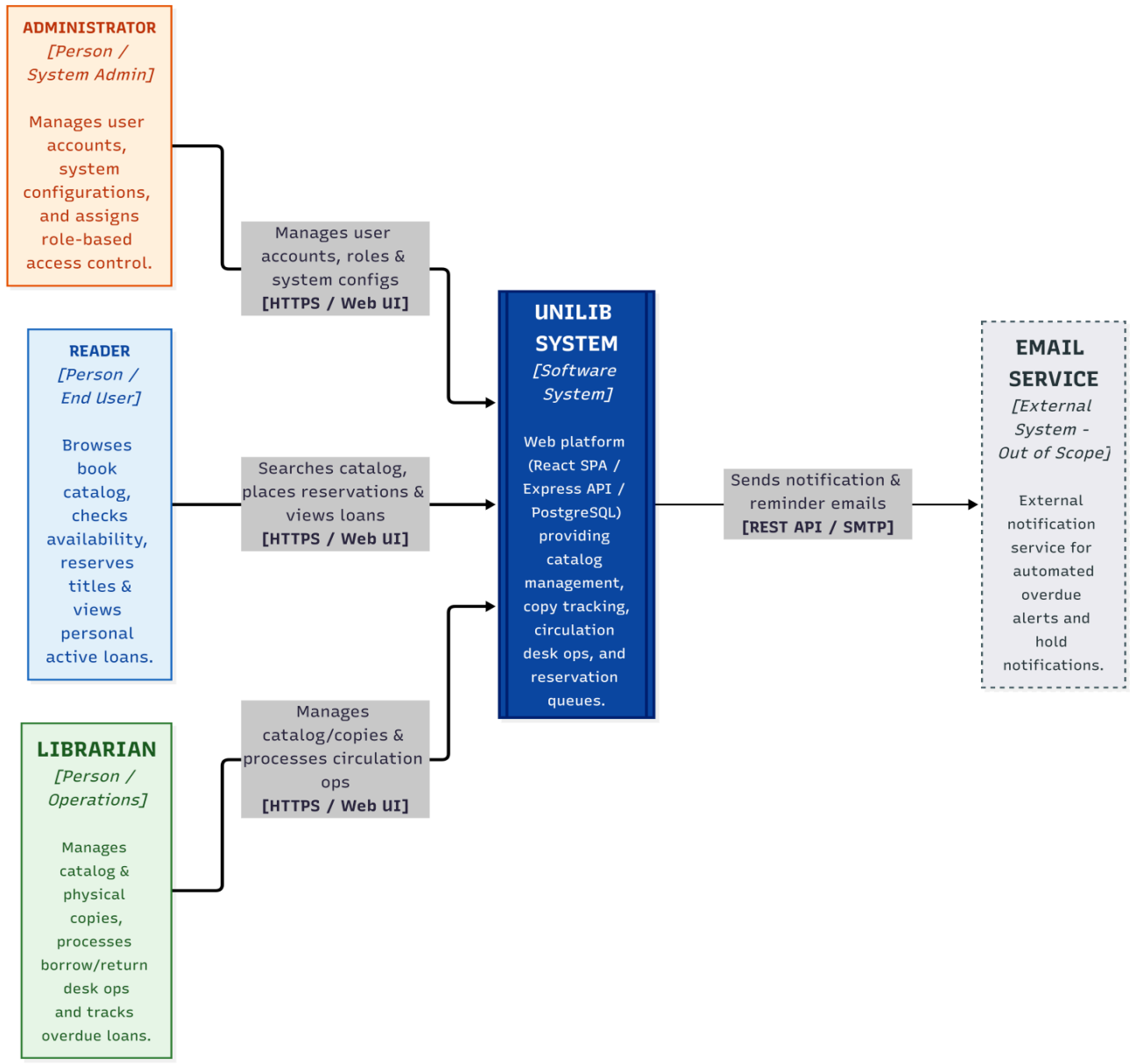
F04	Search & Browse	Multi-attribute catalog search by title, author, category, or ISBN with filtering, pagination, and real-time physical copy availability tags for Readers.	Must
F05	Borrow & Return Management	Circulation workflow handling book checkout, automated due-date calculation (14 days), return processing, copy status updates, and overdue tracking	Must
F06	Book Reservation	FIFO queue-based hold placement for Readers when all physical copies of a title are checked out, with automatic status updates upon copy return.	Must
F07	Reader Overdue & Loan Dashboard	Visual dashboard displaying active student loans, upcoming due dates, overdue notices, and current reservation queue status.	should

Out of scope:

- E-book & Digital Rights Management (DRM): No digital reading features, PDF/ePub hosting, or digital copyright control.
- Online Payment Gateway & Fine Processing: No online fine calculation or integration with payment platforms (MoMo, VNPay, credit cards).
- Hardware Barcode / RFID Integration: No physical hardware scanner drivers or RFID gate reader integration; copy IDs are managed via standard text/manual input.
- Inter-library Loan (ILL) & Distributed Operations: No cross-library book transfers, external library API synchronization, or multi-branch operations.
- AI-based Recommendation Engine: No automated reading recommendations or personalized book suggestion algorithms.
- SMS / Production Email Notifications: System will not send actual SMS messages or automated production emails for overdue alerts or reservation pickups.
- Native Mobile Applications: Dedicated Android/iOS mobile apps are excluded; access is strictly via responsive web browser.

3 Proposed Solution

3.1 Preliminary Architecture



3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	React with JavaScript and Vite	Reusable components suit the Reader, Librarian, and Administrator screens. Vite provides a fast React development/build workflow; a browser-based application matches PA0 without a native mobile client.
Back end	Node.js with Express, REST/JSON API	JavaScript across client and server reduces language switching for the student team. Express routing and middleware support the six feature modules, input validation, authorization, and consistent error handling.
Database	PostgreSQL	Relational constraints and transactions suit users, book titles, physical copies, loans, and reservations. Borrow/return operations need coordinated atomic updates; detailed schema and concurrency rules belong to PA3.
Hosting / deployment	Vercel for the React frontend; Render for the Express API and PostgreSQL	Retains the PA0 proposal and separates static frontend delivery from the API/database. Hosting is planned for PA5; budget, database retention, and service limits must be checked before provisioning.
Testing & CI	Vitest + React Testing Library for frontend tests; Jest + Supertest for backend tests; GitHub Actions for CI	Matches WBS 4.1.2: component tests cover user interactions, API tests cover authorization and circulation rules, and CI provides repeatable checks. PA1 currently runs repository checks only; test setup is scheduled for PA4 and application test execution for PA5.

3.3 Hardware and Operating Environment

[Describe the hardware, devices and environments needed to develop, test and run the software.]

- Development hardware: Each team member needs a laptop or desktop computer capable of running a modern web browser, Node.js, npm, Git, and a code editor such as Visual Studio Code. No specialized hardware is required.
- Operating system: Windows, macOS, or Linux may be used, provided the required development tools and a supported Node.js version are installed.

- Browser and client environment: Use an up-to-date version of Google Chrome, Mozilla Firefox, or Microsoft Edge to access and test the UniLib web application.
- Local development environment: Install Node.js 22 LTS and npm. Use these to run the React + Vite frontend and the Node.js + Express backend locally. Use Git and the team GitHub repository for version control and collaboration.
- Testing environment: Test the frontend and backend locally with the agreed testing tools (Vitest and React Testing Library for the frontend; Jest and Supertest for the backend). A separate physical test device is not required for the planned web application.
- Deployment and runtime environment: The planned deployment uses Vercel for the frontend and Render for the Express API and PostgreSQL database, subject to free-tier limits and the team's final deployment decision. The team should prepare seed data and a demo plan in case a free database or service has availability or retention limits.
- Network access: An internet connection is needed to access GitHub, install dependencies, collaborate, and use the hosted application. Local development and testing can be performed on each member's computer.

4 Project Plan

4.1 Process and Phase Gates

Decision status: Proposal prepared by Quân for PA1-Q2; review by Ánh and team confirmation are pending. This preserves PA0's proposed core stack and F01–F06 scope. No application packages, detailed schema, or deployment are introduced in PA1.

Implementation constraints: Select supported, mutually compatible versions when the application is initialized and record them in manifests/lockfiles. The current repository CI uses Node.js 22; Vite currently requires Node.js 20.19+ or 22.12+, so verify the chosen toolchain before setup. Render's free web services can spin down after inactivity and free PostgreSQL databases expire after 30 days; Vinh must include a viable demo/data-retention plan in the cost estimate rather than assume free hosting lasts the entire course.

References: Vite setup, Express middleware, PostgreSQL transactions, Vercel Vite deployment, Render free-service limits, React Testing Library, Supertest.

PA	Phase	Due	Exit criteria (fill in)
PA1	Initiation & planning	End of week 3	• Project Proposal & Plan approved and submitted (Vision, Scope, Solution architecture overview, 3-level WBS, Schedule/Gantt baseline, Roles & Responsibilities, Risk Management). • Git repository structure (/src, /docs, /pa) initialized; main branch protection configured; GitHub Projects task board populated with all WBS work packages. • Baseline agreed by all 4 members and verified compliant with PA0 constraints.
PA2	Requirements	End of week 5	• Software Requirements Specification (SRS) completed and baselined, specifying functional requirements for all 6 Must-have features (F01–F06) and non-functional requirements.

			• Complete Use Case specifications and Use Case diagram delivered for Reader, Librarian, and Administrator actors. • Low-fidelity UI wireframes for all core user workflows reviewed and approved by BA and PM. • Requirements Traceability Matrix (RTM) established mapping business needs to functional requirements.
PA3	Design	End of week 7	• Software Architecture Document (SAD) completed with C4 System Context and Container diagrams. • Relational database schema (PostgreSQL) designed with 3NF normalized tables, constraints, and ERD for Book Titles, Copies, Loans, Reservations, and Users/Roles. • RESTful API specification (endpoints, HTTP methods, request/response JSON schemas, error codes) documented and reviewed. • High-fidelity UI mockups and design tokens finalized; Architectural Decision Records (ADRs) recorded.
PA4	Test planning	End of week 8	• Master Test Plan finalized defining test scope, test levels, defect classification, and environment criteria. • Detailed test cases designed for all 6 Must-have features (F01–F06), covering positive paths, boundary conditions, and state transitions (Loans/Reservations). • Requirements-to-test traceability matrix completed with 100% Must-have coverage. • Automated test framework setup and verified in repository.
PA5	Implementation, testing & deployment	End of week 10	• All 6 Must-have features (F01–F06) implemented in React (FE) and Express/PostgreSQL (BE) with passing unit and integration tests. • All code merged to main via pull requests with required peer review approvals and conventional commit history.

			· Full system test execution completed with 0 critical or high-severity open defects; Test Summary Report published. · Web application successfully deployed to cloud hosting (Vercel for FE, Render for BE/DB) with verified staging/production smoke tests.
Final	Presentation & demo	Week 11	· End-to-end live demonstration of all 6 Must-have library workflows executed smoothly without critical failure. · Final presentation slide deck delivered and rehearsed by all team members. · Final project repository, documentation baselines, individual contribution assessments, and project retrospective submitted.

4.2 Work Breakdown Structure (WBS)

The UniLib project is structured using a hierarchical Work Breakdown Structure (WBS) with 3 levels of decomposition:

· Level 1 (Phase): Aligned with the project life cycle gates (PA1 to Final).

· Level 2 (Deliverable / Sub-phase): Major milestones or functional groupings.

· Level 3 (Work Package): Atomic, manageable work packages sized between 2 and 8 person-hours, each assigned to exactly one owner and trackable as an issue on GitHub Projects.

All 6 Must-have features (F01 Authentication & Roles, F02 Book Catalog Management, F03 Book Copy Management, F04 Search & Browse, F05 Borrow & Return, F06 Book Reservation) are systematically addressed across PA2 through PA5.

WBS Master Dictionary Table

WBS ID	Work Package Name	Description & Key Deliverables	Owner	Effort (h)	Stream / Feature
1.0	**PA1 Initiation & Planning**	**Project kickoff, baseline scope, planning document, and tooling**	**Vinh**	**54**	**Management**
1.1.1	Project Kickoff Meeting & Charter Sign-off	Conduct kickoff meeting, agree on working rules, internal deadlines,	Vinh	3	Management

		and finalize kickoff meeting minutes.			
1.1.2	Governance & Roles Alignment	Formalize main/backup role assignments, conflict resolution workflow, and decision-making protocol.	Vinh	2	Management
1.2.1	Preliminary Vision & Problem Statement	Draft business problem description, target user profiles, and product positioning statement.	Nguyên	5	Vision
1.2.2	Stakeholder Needs & MoSCoW Scope	Define stakeholder profiles and categorize features into MoSCoW priority, strictly establishing out-of-scope boundaries.	Nguyên	4	Scope
1.2.3	Preliminary Architecture & Technology Justification	Construct preliminary C4 context diagram and evaluate stack suitability (React, Express, PostgreSQL, Vercel/Render).	Ánh	4	Architecture
1.2.4	Hardware & Operating Environment Specification	Specify developer machines, browser support targets, and cloud hosting constraints.	Ánh	3	Architecture
1.2.5	Process Gates & Exit Criteria Definition	Define phase gate milestones (PA1 to Final) with specific, verifiable exit criteria.	Vinh	3	Planning
1.2.6	Work Breakdown Structure (WBS) Authoring	Structure hierarchical 3-level WBS (phases, deliverables, work packages of 2–8h) with single ownership.	Vinh	6	Planning
1.2.7	Effort & Cost Estimation Baseline	Calculate person-hour estimates and formulate human resource/tooling cost model.	Vinh	4	Planning
1.2.8	Project Schedule & Critical Path Mapping	Build schedule network, identify milestone dependencies, critical path, and initial Gantt chart.	Ánh	5	Planning
1.2.9	Project Risk Identification & Mitigation	Identify at least 5 key risks across technical, people, and organizational categories; calculate	Vinh	4	Risk

		RE (P × L) and define contingency plans.			
1.3.1	Version Control & Repository Structure Setup	Initialize GitHub repository with standard structure (`/src`, `/docs`, `/pa`), README, .gitignore, and branch protection rules.	Quân	3	Infrastructure
1.3.2	Task Board & Tracking Configuration	Set up GitHub Projects board, configure kanban columns, labels, milestones, and import initial WBS tasks.	Quân	4	Infrastructure
1.3.3	PA1 Document Compilation & Review	Consolidate individual contributions into PA1 markdown, perform peer review, and finalize submission baseline.	Vinh	4	Management
2.0	**PA2 Requirements Engineering**	**Software Requirements Specification (SRS), Use Cases, Wireframes & RTM**	**Nguyên**	**60**	**Requirements**
2.1.1	Domain Modeling & Glossary Definition	Analyze library circulation domain; define standardized glossary (Book Title, Book Copy, Loan, Reservation, Overdue).	Nguyên	4	Domain
2.1.2	System Use Case Modeling	Construct system-level Use Case diagram for Reader, Librarian, and Administrator actors.	Nguyên	4	Domain
2.2.1	F01 User Authentication & Roles Specification	Detail functional requirements, login flows, role-based authorization rules (Reader, Librarian, Admin), and use case specs.	Nguyên	4	F01 Auth
2.2.2	F02 Book Catalog Management Specification	Specify functional requirements and use cases for adding, editing, and viewing book titles, ISBNs, authors, and categories.	Nguyên	5	F02 Catalog
2.2.3	F03 Book Copy Management Specification	Specify requirements for tracking physical copies, copy IDs/barcodes, and Available/Borrowed/Reserved state transitions.	Nguyên	5	F03 Copy

2.2.4	F04 Search & Browse Specification	Specify search queries (title, author, category, ISBN), multi-attribute filtering, and real-time availability display.	Nguyên	4	F04 Search
2.2.5	F05 Borrow & Return Management Specification	Detail use case flows for loan checkout, due date calculation, return processing, copy state updates, and overdue tracking rules.	Nguyên	6	F05 Loan
2.2.6	F06 Book Reservation Specification	Specify reservation queue rules, title reservation when all copies are borrowed, reservation priority, and pickup hold states.	Nguyên	5	F06 Reservation
2.2.7	Non-Functional Requirements & Constraints	Detail performance, security, data integrity, usability, and browser support requirements.	Ánh	4	NFR
2.3.1	Reader/Student Workflow Wireframes	Create UI wireframes for search/browse, book detail, reservation placement, and personal loan/reservation views.	Ánh	5	UI/UX
2.3.2	Librarian & Admin Management Wireframes	Create UI wireframes for catalog CRUD, copy inventory, circulation desk (borrow/return/overdue), and user administration.	Ánh	6	UI/UX
2.4.1	Requirements Traceability Matrix (RTM) Baseline	Construct RTM linking user needs to functional requirements (F01–F06) and establish test verification methods.	Vinh	4	QA / Traceability
2.4.2	PA2 Document Review & Verification	Conduct team walkthrough of SRS, resolve requirement ambiguities, verify against PA1 baseline, and submit PA2.	Vinh	4	Management
3.0	**PA3 Architecture & Detailed Design**	**System Architecture (C4), Relational Schema (PostgreSQL), REST API & UI Mockups**	**Ánh**	**56**	**Design**
3.1.1	C4 Architecture Modeling (Context & Container)	Create C4 Context and Container diagrams specifying React SPA,	Ánh	5	Architecture

		Node.js/Express REST API, and PostgreSQL DB.			
3.1.2	Architectural Decision Records (ADRs)	Author ADRs for JWT authentication, database access patterns, API error handling, and state management strategy.	Quân	4	Architecture
3.2.1	Conceptual & Logical Data Modeling (ERD)	Design Entity-Relationship Diagram covering Users, Roles, BookTitles, BookCopies, Loans, and Reservations.	Ánh	5	Database
3.2.2	Physical Database Schema & DDL Scripts	Write normalized 3NF PostgreSQL DDL schema with primary/foreign keys, indexes, check constraints, and seed data scripts.	Quân	5	Database
3.3.1	API Specification for F01 Auth & User Management	Specify RESTful API endpoints, request/response JSON schemas, and HTTP status codes for auth and users.	Quân	4	F01 Auth
3.3.2	API Specification for F02 Catalog & F03 Copy Management	Specify REST endpoints for book titles CRUD, copy additions, and inventory queries.	Quân	5	F02 & F03
3.3.3	API Specification for F04 Search & Browse	Specify endpoints for catalog search with query parameters, filtering, pagination, and copy counts.	Quân	4	F04 Search
3.3.4	API Specification for F05 Borrow & Return	Specify endpoints for loan checkout, check-in, loan renewal/due date, and overdue loan listings.	Quân	5	F05 Loan
3.3.5	API Specification for F06 Book Reservation	Specify endpoints for creating reservations, viewing queue status, cancellation, and pickup fulfillment.	Quân	4	F06 Reservation
3.4.1	UI Design System & Component Library	Design reusable UI components (buttons, modals, tables, badges, form inputs), color scheme, and typography tokens.	Ánh	5	UI/UX

3.4.2	High-Fidelity Mockups for Key Workflows	Produce high-fidelity mockups for Reader search/reserve and Librarian circulation desk.	Ánh	6	UI/UX
3.5.1	Design Review & Traceability Verification	Inspect API and DB designs against SRS functional requirements; update RTM and compile PA3 deliverable.	Vinh	4	Management
4.0	**PA4 Test Planning & QA Setup**	**Master Test Plan, Test Cases for F01–F06, Test Environment & Automated Setup**	**Vinh**	**41**	**Testing**
4.1.1	Master Test Plan Authoring	Define test strategy, test levels (unit, integration, system), environment configuration, entry/exit criteria, and defect severity levels.	Vinh	5	QA Strategy
4.1.2	Test Automation Framework Setup	Configure backend testing tools (Jest, Supertest) and frontend testing tools (Vitest/React Testing Library) with sample test cases.	Quân	4	QA Setup
4.2.1	Test Cases for F01 Authentication & Roles	Author detailed test cases for login validation, password hashing, role-based route guards, and token expiry.	Vinh	5	F01 Auth
4.2.2	Test Cases for F02 Catalog & F03 Copy Management	Author test cases for book title CRUD, ISBN uniqueness, copy status updates (Available/Borrowed/Reserved), and deletions.	Vinh	5	F02 & F03
4.2.3	Test Cases for F04 Search & Browse	Author test cases for keyword search, filtering by category/author, pagination, and availability counter accuracy.	Vinh	4	F04 Search
4.2.4	Test Cases for F05 Borrow & Return	Author test cases for loan creation, due date setting, return processing, copy status transitions, and overdue loan detection.	Vinh	6	F05 Loan
4.2.5	Test Cases for F06 Book Reservation	Author test cases for reserving unavailable books, reservation	Vinh	5	F06 Reservation

		queue FIFO, hold status on return, and user cancellation.			
4.3.1	Test Traceability Matrix & Test Data Specification	Map test cases to SRS requirements (F01–F06) in RTM and document mock test datasets.	Nguyên	4	QA / Traceability
4.3.2	PA4 Document Review & Baseline	Review test cases for completeness with Developer and BA; finalize and submit PA4.	Vinh	3	Management
5.0	**PA5 Implementation, Testing & Deployment**	**Frontend (React), Backend (Express), PostgreSQL, Integration & Cloud Deployment**	**Quân**	**91**	**Development**
5.1.1	Backend Boilerplate & DB Connection Setup	Initialize Express app, configure PostgreSQL connection pool, migration scripts, and error-handling middleware.	Quân	5	Dev Infrastructure
5.1.2	Frontend Boilerplate & Layout Setup	Initialize React app, configure router, global state/context for auth, and common layout shell (navbar, sidebar).	Quân	5	Dev Infrastructure
5.2.1	F01 Authentication & Role-Based Access Control	Implement JWT auth endpoints, password hashing (bcrypt), login UI, and role-protected routes (Reader, Librarian, Admin).	Quân	6	F01 Auth
5.2.2	F02 Book Catalog Management Module	Implement backend CRUD endpoints and Librarian UI to add, edit, and delete book titles.	Quân	6	F02 Catalog
5.2.3	F03 Book Copy Management Module	Implement backend copy endpoints and UI to register physical copies with barcode IDs and manage Available/Borrowed status.	Quân	5	F03 Copy
5.2.4	F04 Search & Browse Books Module	Implement catalog search API with multi-attribute filtering and responsive Reader UI with availability tags.	Ánh	6	F04 Search

5.2.5	F05 Borrow & Return Management Module	Implement circulation API/UI for loan checkout, return check-in, copy status sync, and Librarian overdue list.	Quân	8	F05 Loan
5.2.6	F06 Book Reservation Module	Implement reservation placement when copies are unavailable, queue ordering, and hold transition on copy return.	Quân	7	F06 Reservation
5.2.7	Reader Dashboard & Active Loans View	Implement Reader dashboard displaying current borrowed books, due dates, overdue notices, and active reservations.	Ánh	5	UI / Reader
5.3.1	Automated Unit & Integration Testing	Implement automated API tests for auth, circulation rules, copy status transitions, and reservation constraints.	Quân	6	QA Automation
5.3.2	Manual System Testing & Bug Logging	Execute PA4 test suite against integrated web app; log bugs with steps and severity on GitHub Issues.	Vinh	7	System Testing
5.3.3	Bug Fixing & Regression Verification	Fix reported defects in frontend and backend; verify resolved issues and perform regression testing.	Quân	6	Bug Fixing
5.3.4	Test Summary Report & QA Sign-off	Compile test execution metrics, defect resolution stats, and finalize Test Summary Report.	Vinh	4	QA Sign-off
5.4.1	Cloud DB & Backend Deployment on Render	Provision PostgreSQL database on Render, run production migrations, deploy Express API, and configure environment variables.	Quân	4	Deployment
5.4.2	Frontend Cloud Deployment on Vercel	Deploy React frontend to Vercel, configure production API base URL, CORS, and custom build scripts.	Quân	4	Deployment
5.4.3	Production Smoke Testing &	Perform live smoke testing across all 6 features on production URLs;	Vinh	4	Deployment

	Deployment Documentation	document deployment instructions in README.			
5.4.4	PA5 Submission & Code Baseline	Review final repository status, verify deployment health, package artifacts, and submit PA5.	Vinh	3	Management
6.0	**Final Presentation, Demo & Closeout**	**End-to-End System Demo, Presentation Slide Deck, Contribution Report & Retrospective**	**Vinh**	**16**	**Final**
6.1.1	Demo Scenario Scripting & Seed Data Preparation	Prepare comprehensive demonstration script and realistic seed data covering all 6 Must-haves and edge cases.	Nguyên	4	Demo
6.1.2	Demo Rehearsal & Backup Recording	Conduct full team dry run of live demonstration and record backup video walkthrough of core workflows.	Vinh	3	Demo
6.2.1	Presentation Slide Deck Creation	Author final presentation slides detailing project vision, architecture, demo, metrics, and teamwork reflection.	Ánh	5	Presentation
6.2.2	Final Contribution Assessment & Project Retrospective	Compile individual contribution evidence, calculate final percentages, document retrospective lessons, and submit final package.	Vinh	4	Closeout
Total	**Project Total Estimated Effort**	**Full project life cycle across 11 weeks (4 members)**	**All**	**318**	**All Streams**

4.3 Estimation

4.3.1 Estimation Methodology & Principles

Effort estimation for the UniLib project is conducted using a bottom-up technique based on the 3-level Work Breakdown Structure (WBS). To ensure rigorous, realistic, and defensible effort allocations, the team applies the Three-Point Estimation technique (PERT) combined with Expert

Judgement and Analogy from previous academic and professional software engineering project data.

For each of the 69 atomic work packages, three estimates are established in person-hours:

- Optimistic (O): Effort required assuming optimal conditions, familiar tools, minimal defect discovery, and immediate peer review turnaround.
- Most Likely (M): Realistic effort expected under normal student working conditions, accounting for typical syntax debugging and testing overhead.
- Pessimistic (P): Effort required under unfavorable scenarios, including subtle asynchronous state bugs, database constraint conflicts, or cloud deployment configuration friction.

The expected person-hour effort (E) is calculated using the standard PERT weighted formula: $E = (O + 4M + P) / 6$.

4.3.2 Key Estimation Assumptions

1. Team Bandwidth & Course Constraints: The project spans 11 academic weeks with 4 team members. Each member commits 6–8 person-hours per week according to the course rules, yielding a total team capacity of 264–352 person-hours.
2. Scope Containment: Strict adherence to the 6 Must-have features (F01–F06) approved in PA0. Out-of-scope capabilities (hardware barcode scanners, payment gateways, mobile apps, e-book readers) require zero effort.
3. Unified JavaScript Stack: Using React (Vite) on the frontend and Node.js (Express) on the backend significantly reduces context-switching penalties between client and server development.
4. Buffer Allocation: Sizable buffers (up to 40% variance between O and P) are allocated to PA5 implementation (91 person-hours), automated integration testing, and cloud deployment on Render/Vercel to guard against student-schedule compression.

4.3.3 WBS Effort Reconciliation & Team Capacity Alignment.

During preliminary proposal drafting, coarse placeholder totals (274h) were tentatively noted. The detailed bottom-up decomposition across all 69 level-3 work packages yields exactly 318

person-hours. Distributed over the 11-week life cycle across 4 members, this represents: 318 hours / (4 members × 11 weeks) = 7.23 person-hours/member/week. This aligns precisely with the 6–8 hours/week capacity commitment, confirming that the project plan is fully achievable without requiring overtime.

WBS ID	Work Package Name	Owner	Effort (h)	Method & Specific Assumptions
PA1	**Phase PA1 Work Packages Subtotal**	**Team**	**54 h**	**Subtotal of all level-3 packages in Phase PA1**
1.1.1	Project Kickoff Meeting & Charter Sign-off	Vinh	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; 4 members present; agenda prepared; 1.25h meeting + 1.75h minutes documentation
1.1.2	Governance & Roles Alignment	Vinh	2 h (O:1, M:2, P:3)	Three-point (PERT) & expert judgement; Formalizing primary/backup role matrix and escalation rules based on Team Contract
1.2.1	Preliminary Vision & Problem Statement	Nguyên	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Analyzing small/university library operational pains; writing business problem statement
1.2.2	Stakeholder Needs & MoSCoW Scope	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; MoSCoW classification fixing F01–F06 as Must-have; confirming strict out-of-scope boundaries
1.2.3	Preliminary Architecture & Technology Justification	Ánh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Evaluating React/Node/Express/PostgreSQL against student skillset and deployment feasibility
1.2.4	Hardware & Operating Environment Specification	Ánh	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; Documenting modern evergreen browser

				targets (Chrome/Firefox/Edge) and Node.js 22 LTS runtime
1.2.5	Process Gates & Exit Criteria Definition	Vinh	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; Establishing rigorous, verifiable exit criteria for PA1 through Final demo
1.2.6	Work Breakdown Structure (WBS) Authoring	Vinh	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Decomposing 6 phases into 69 atomic packages (2–8h) with single ownership
1.2.7	Effort & Cost Estimation Baseline	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Applying PERT three-point estimation; reconciling 318h against 6–8h/wk capacity; costing junior rate
1.2.8	Project Schedule & Critical Path Mapping	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Mapping calendar dependencies, determining critical path, scheduling PA milestones
1.2.9	Project Risk Identification & Mitigation	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Assessing 5+ risk scenarios across tech, people, and schedule; calculating Risk Exposure (RE = P x L)
1.3.1	Version Control & Repository Structure Setup	Quân	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; Git init with /src, /docs, /pa skeletons, .gitignore, branch protection rules for main
1.3.2	Task Board & Tracking Configuration	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; GitHub Projects board setup with Backlog/Todo/In Progress/Review/Done

				columns and issue mapping
1.3.3	PA1 Document Compilation & Review	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Consolidating all contributions, performing team review, verifying formatting and baseline submission
PA2	**Phase PA2 Work Packages Subtotal**	**Team**	**60 h**	**Subtotal of all level-3 packages in Phase PA2**
2.1.1	Domain Modeling & Glossary Definition	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Domain analysis for circulation; standardizing terms: Book Title, Book Copy, Loan, Reservation, Overdue
2.1.2	System Use Case Modeling	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; UML Use Case diagram modeling Reader, Librarian, and Administrator boundaries
2.2.1	F01 User Authentication & Roles Specification	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Specifying authentication, credential validation, RBAC route guards for 3 roles
2.2.2	F02 Book Catalog Management Specification	Nguyên	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Specifying book title catalog metadata (title, author, ISBN, category, publisher, year)
2.2.3	F03 Book Copy Management Specification	Nguyên	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Specifying physical copy tracking with unique copy barcode/ID and Available/Borrowed/Reserved states
2.2.4	F04 Search & Browse Specification	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement;

				Specifying keyword search, multi-field filters, pagination, and real-time copy availability counter
2.2.5	F05 Borrow & Return Management Specification	Nguyên	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Circulation logic: loan checkout, return check-in, loan period calculation, overdue detection rules
2.2.6	F06 Book Reservation Specification	Nguyên	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; FIFO reservation queue specification; auto-hold assignment on copy return; reservation cancellation
2.2.7	Non-Functional Requirements & Constraints	Ánh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Specifying response time (<2s), concurrent user targets (20), data integrity, security constraints
2.3.1	Reader/Student Workflow Wireframes	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Low-fidelity wireframes for Reader search, catalog view, title reservation, and my-loans dashboard
2.3.2	Librarian & Admin Management Wireframes	Ánh	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Low-fidelity wireframes for Librarian catalog CRUD, physical copy inventory, circulation desk, overdue list
2.4.1	Requirements Traceability Matrix (RTM) Baseline	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Mapping stakeholder needs to functional requirements F01–F06 and test verification methods
2.4.2	PA2 Document Review & Verification	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Team

				walkthrough of SRS, resolving requirement gaps, verifying compliance with PA1 baseline
PA3	**Phase PA3 Work Packages Subtotal**	**Team**	**56 h**	**Subtotal of all level-3 packages in Phase PA3**
3.1.1	C4 Architecture Modeling (Context & Container)	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; C4 System Context & Container diagrams defining React SPA, Express REST API, and PostgreSQL DB
3.1.2	Architectural Decision Records (ADRs)	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Documenting ADRs for JWT auth, PostgreSQL connection pool, REST API conventions, error handling
3.2.1	Conceptual & Logical Data Modeling (ERD)	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Conceptual and logical ERD covering Users, Roles, BookTitles, BookCopies, Loans, Reservations
3.2.2	Physical Database Schema & DDL Scripts	Quân	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Normalized 3NF PostgreSQL DDL script with foreign keys, check constraints, unique ISBN/copy IDs, seed data
3.3.1	API Specification for F01 Auth & User Management	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Specifying REST endpoints /api/auth/login, /api/auth/me, /api/users with JSON schemas and status codes
3.3.2	API Specification for F02 Catalog & F03 Copy Management	Quân	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Specifying REST endpoints for book titles

				CRUD (/api/books) and copy inventory (/api/books/:id/copies)
3.3.3	API Specification for F04 Search & Browse	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Specifying query parameters (q, category, author, isbn, page, limit) and response schema with copy availability
3.3.4	API Specification for F05 Borrow & Return	Quân	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Specifying circulation endpoints (/api/loans/checkout, /api/loans/:id/return, /api/loans/overdue) with atomic updates
3.3.5	API Specification for F06 Book Reservation	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Specifying reservation endpoints (/api/reservations, /api/reservations/:id/cancel) and queue transition specs
3.4.1	UI Design System & Component Library	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Establishing UI design tokens (color palette, typography, spacing) and reusable component specs in Figma
3.4.2	High-Fidelity Mockups for Key Workflows	Ánh	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; High-fidelity mockups for key circulation and catalog workflows for Reader, Librarian, Admin
3.5.1	Design Review & Traceability Verification	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Traceability check linking API/DB design to SRS; consolidating Software Architecture Document (SAD)
PA4	**Phase PA4 Work Packages Subtotal**	**Team**	**41 h**	**Subtotal of all level-3 packages in Phase PA4**

4.1.1	Master Test Plan Authoring	Vinh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Authoring Master Test Plan: defining unit, integration, system test levels, entry/exit criteria, bug severity
4.1.2	Test Automation Framework Setup	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Configuring Jest + Supertest (backend) and Vitest + React Testing Library (frontend) config files in repo
4.2.1	Test Cases for F01 Authentication & Roles	Vinh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Designing test cases for auth: valid login, wrong password, role route access, expired token
4.2.2	Test Cases for F02 Catalog & F03 Copy Management	Vinh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Designing test cases for catalog CRUD: duplicate ISBN, mandatory fields, copy addition, state validation
4.2.3	Test Cases for F04 Search & Browse	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Designing test cases for catalog search: query matches, multi-filter, pagination, availability display
4.2.4	Test Cases for F05 Borrow & Return	Vinh	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Designing test cases for loan checkout (available copy only), due date calculation, return check-in, overdue tracking
4.2.5	Test Cases for F06 Book Reservation	Vinh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Designing test cases for reservation: placing hold on 0 available copies, FIFO queue order, hold status on return

4.3.1	Test Traceability Matrix & Test Data Specification	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Updating RTM to ensure 100% test case coverage for F01–F06 and defining mock test fixtures
4.3.2	PA4 Document Review & Baseline	Vinh	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; Team inspection of test cases, cross-verification with Developer, compiling and submitting PA4 deliverable
PA5	**Phase PA5 Work Packages Subtotal**	**Team**	**91 h**	**Subtotal of all level-3 packages in Phase PA5**
5.1.1	Backend Boilerplate & DB Connection Setup	Quân	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Express server setup, node-postgres connection pool, migration scripts, error handling middleware
5.1.2	Frontend Boilerplate & Layout Setup	Quân	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; React 18 + Vite setup, react-router-dom, AuthContext, UI layout shell (navbar, footer, sidebar)
5.2.1	F01 Authentication & Role-Based Access Control	Quân	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Implementing JWT auth backend (bcrypt hashing, token generation) and React login form with route guards
5.2.2	F02 Book Catalog Management Module	Quân	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Implementing book title CRUD API and Librarian catalog management UI tables/forms
5.2.3	F03 Book Copy Management Module	Quân	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Implementing physical copy registration API and UI with copy

				barcode and Available/Borrowed status
5.2.4	F04 Search & Browse Books Module	Ánh	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Implementing search API with SQL ILIKE/indexes and Reader catalog browsing UI with real-time availability badges
5.2.5	F05 Borrow & Return Management Module	Quân	8 h (O:6, M:8, P:10)	Three-point (PERT) & expert judgement; Implementing circulation API/UI: loan creation with transaction, return check-in, overdue tracking list
5.2.6	F06 Book Reservation Module	Quân	7 h (O:5, M:7, P:9)	Three-point (PERT) & expert judgement; Implementing reservation queue logic in SQL transaction, hold state transition on copy return, reservation cancellation
5.2.7	Reader Dashboard & Active Loans View	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Implementing Reader personal dashboard: active loans list, due dates, overdue notices, and reservation queue status
5.3.1	Automated Unit & Integration Testing	Quân	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Writing automated integration tests for circulation rules, copy status transitions, reservation constraints via Supertest
5.3.2	Manual System Testing & Bug Logging	Vinh	7 h (O:5, M:7, P:9)	Three-point (PERT) & expert judgement; Executing PA4 test suite against integrated app; logging defects with

				reproduction steps on GitHub Issues
5.3.3	Bug Fixing & Regression Verification	Quân	6 h (O:4, M:6, P:8)	Three-point (PERT) & expert judgement; Fixing reported defects in React UI and Express backend; running regression test suite to verify 0 critical bugs
5.3.4	Test Summary Report & QA Sign-off	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Compiling test pass rate metrics, defect resolution stats, and publishing Test Summary Report
5.4.1	Cloud DB & Backend Deployment on Render	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Provisioning PostgreSQL on Render, running DDL migration scripts, deploying Express API web service
5.4.2	Frontend Cloud Deployment on Vercel	Quân	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Deploying React frontend to Vercel, configuring production environment variables (VITE_API_URL) and CORS
5.4.3	Production Smoke Testing & Deployment Documentation	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Executing live production smoke tests across all 6 Must-haves; documenting deployment steps in README
5.4.4	PA5 Submission & Code Baseline	Vinh	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; Final repository check, verifying live deployed links, tagging PA5 release, submitting PA5 deliverable
Final	**Phase Final Work Packages Subtotal**	**Team**	**16 h**	**Subtotal of all level-3 packages in Phase Final**

6.1.1	Demo Scenario Scripting & Seed Data Preparation	Nguyên	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Writing detailed demo script covering end-to-end library workflow and seeding realistic mock library data
6.1.2	Demo Rehearsal & Backup Recording	Vinh	3 h (O:2, M:3, P:4)	Three-point (PERT) & expert judgement; Dry run demo with all 4 members; recording backup video demonstration to mitigate live presentation risks
6.2.1	Presentation Slide Deck Creation	Ánh	5 h (O:3, M:5, P:7)	Three-point (PERT) & expert judgement; Creating slide deck: problem, architecture, demo walkthrough, test results, lessons learned
6.2.2	Final Contribution Assessment & Project Retrospective	Vinh	4 h (O:3, M:4, P:5)	Three-point (PERT) & expert judgement; Compiling final contribution assessment (25% each), team retrospective report, and final course closeout
Total	**Project Total Estimated Effort**	**All Members**	**318 h**	**Complete 11-week lifecycle effort: 7.23 h/person/week average across 4 members**

4.4 Schedule

4.4.1 Schedule Architecture:

Effort versus Duration: a fundamental project management principle applied in UniLib is the explicit distinction between Work Effort and Task Duration:

· Effort (person-hours): The actual net labor time required to execute the deliverable (totaling 318 person-hours across all 69 packages).

· Duration (calendar days/weeks): The elapsed calendar time from task inception to completion. Because team members are university students balancing other coursework with an allocation of 6–8 hours/week, a 6-hour work package typically spans 3–4 calendar days.

The project follows a linear Waterfall progression structured into 6 major lifecycle phases and anchored by strict Phase Gate Milestones. Work package dependencies are modeled using Finish-to-Start (FS) logic with predecessor constraints.

4.4.2 Project Phase Gate Milestones Table

Milestone	Phase	Key Deliverables & Exit Criteria	Target Baseline Date	Critical Path?
M1	PA1	Project Proposal & Plan baselined, repo structure protected, GitHub Projects board populated with 69 WBS tasks.	2026-10-21 (Week 1)	**Yes [CP]**
M2	PA2	SRS completed with detailed specs for F01–F06, Use Case model, wireframes, and RTM baseline.	2026-11-04 (Week 3)	**Yes [CP]**
M3	PA3	C4 architecture (Context/Container), 3NF PostgreSQL schema DDL, REST API specs for F01–F06, and UI mockups.	2026-11-18 (Week 5)	**Yes [CP]**
M4	PA4	Master Test Plan, detailed test cases for F01–F06 with 100% requirements coverage, and automated test setup.	2026-11-25 (Week 6)	**Yes [CP]**
M5	PA5	Full implementation of F01–F06, automated test suite passing, 100% critical bugs resolved, deployed to Vercel/Render.	2026-12-09 (Week 8)	**Yes [CP]**
M6	Final	Live system demo of all 6 Must-haves, final slide deck, peer contribution assessments, and project retrospective.	2026-12-16 (Week 9)	**Yes [CP]**

4.4.3 Master Project Schedule & Gantt Data Table

	Task / Work Package Name	Owner	Effort	Duration	Start Date	End Date	Predecessors	CP?
PA1	**Phase PA1 Timeline**	**Team**	**54h**	6d	2026-10-15	2026-10-21	**-**	**[CP]**
1.1.1	Project Kickoff Meeting & Charter Sign-off	Vinh	3h	0d	2026-10-15	2026-10-15	None	**[CP]**
1.1.2	Governance & Roles Alignment	Vinh	2h	1d	2026-10-15	2026-10-16	1.1.1	-
1.2.1	Preliminary Vision & Problem Statement	Nguyên	5h	2d	2026-10-16	2026-10-18	1.1.1	**[CP]**
1.2.2	Stakeholder Needs & MoSCoW Scope	Nguyên	4h	2d	2026-10-17	2026-10-19	1.2.1	**[CP]**
1.2.3	Preliminary Architecture & Technology Justification	Ánh	4h	2d	2026-10-16	2026-10-18	1.2.1	-
1.2.4	Hardware & Operating Environment Specification	Ánh	3h	2d	2026-10-17	2026-10-19	1.2.3	-
1.2.5	Process Gates & Exit	Vinh	3h	1d	2026-10-16	2026-10-17	1.1.2	-

	Criteria Definition							
1.2.6	Work Breakdown Structure (WBS) Authoring	Vinh	6h	2d	2026-10-17	2026-10-19	1.2.2, 1.2.5	**[CP]**
1.2.7	Effort & Cost Estimation Baseline	Vinh	4h	2d	2026-10-18	2026-10-20	1.2.3, 1.2.6	–
1.2.8	Project Schedule & Critical Path Mapping	Ánh	5h	2d	2026-10-18	2026-10-20	1.2.6, 1.2.7	**[CP]**
1.2.9	Project Risk Identification & Mitigation	Vinh	4h	1d	2026-10-19	2026-10-20	1.2.6, 1.2.7	–
1.3.1	Version Control & Repository Structure Setup	Quân	3h	1d	2026-10-16	2026-10-17	1.1.1	–
1.3.2	Task Board & Tracking Configuration	Quân	4h	2d	2026-10-18	2026-10-20	1.2.6, 1.3.1	–
1.3.3	PA1 Document Compilation & Review	Vinh	4h	1d	2026-10-20	2026-10-21	1.2.4, 1.2.8, 1.2.9, 1.3.2	**[CP]**
PA2	**Phase PA2 Timeline**	**Team**	**60h**	13d	2026-10-22	2026-11-04	–	**[CP]**

2.1.1	Domain Modeling & Glossary Definition	Nguyên	4h	2d	2026-10-22	2026-10-24	M1 (PA1)	**[CP]**
2.1.2	System Use Case Modeling	Nguyên	4h	3d	2026-10-24	2026-10-27	2.1.1	**[CP]**
2.2.1	F01 User Authenticatio n & Roles Specification	Nguyên	4h	3d	2026-10-27	2026-10-30	2.1.2	-
2.2.2	F02 Book Catalog Management Specification	Nguyên	5h	3d	2026-10-27	2026-10-30	2.1.2	-
2.2.3	F03 Book Copy Management Specification	Nguyên	5h	3d	2026-10-28	2026-10-31	2.2.2	-
2.2.4	F04 Search & Browse Specification	Nguyên	4h	3d	2026-10-28	2026-10-31	2.2.2, 2.2.3	-
2.2.5	F05 Borrow & Return Management Specification	Nguyên	6h	3d	2026-10-29	2026-11-01	2.2.3	**[CP]**
2.2.6	F06 Book Reservation Specification	Nguyên	5h	3d	2026-10-30	2026-11-02	2.2.5	**[CP]**
2.2.7	Non-Functio nal Requirement	Ánh	4h	3d	2026-10-29	2026-11-01	2.1.2	-

	s & Constraints							
2.3.1	Reader/Student Workflow Wireframes	Ánh	5h	3d	2026-10-30	2026-11-02	2.2.4, 2.2.6	-
2.3.2	Librarian & Admin Management Wireframes	Ánh	6h	3d	2026-10-30	2026-11-02	2.2.2, 2.2.3, 2.2.5	-
2.4.1	Requirements Traceability Matrix (RTM) Baseline	Vinh	4h	2d	2026-11-01	2026-11-03	2.2.1, 2.2.2, 2.2.3, 2.2.4, 2.2.5, 2.2.6	**[CP]**
2.4.2	PA2 Document Review & Verification	Vinh	4h	2d	2026-11-02	2026-11-04	2.3.1, 2.3.2, 2.4.1	**[CP]**
PA3	**Phase PA3 Timeline**	**Team**	**56h**	13d	2026-11-05	2026-11-18	**-**	**[CP]**
3.1.1	C4 Architecture Modeling (Context & Container)	Ánh	5h	3d	2026-11-05	2026-11-08	M2 (PA2)	**[CP]**
3.1.2	Architectural Decision Records (ADRs)	Quân	4h	3d	2026-11-07	2026-11-10	3.1.1	-

3.2.1	Conceptual & Logical Data Modeling (ERD)	Ánh	5h	3d	2026-11-07	2026-11-10	3.1.1	**[CP]**
3.2.2	Physical Database Schema & DDL Scripts	Quân	5h	3d	2026-11-10	2026-11-13	3.2.1	**[CP]**
3.3.1	API Specification for F01 Auth & User Management	Quân	4h	3d	2026-11-10	2026-11-13	3.1.1, 3.2.1	-
3.3.2	API Specification for F02 Catalog & F03 Copy Management	Quân	5h	3d	2026-11-11	2026-11-14	3.2.2	-
3.3.3	API Specification for F04 Search & Browse	Quân	4h	3d	2026-11-12	2026-11-15	3.3.2	-
3.3.4	API Specification for F05 Borrow & Return	Quân	5h	3d	2026-11-12	2026-11-15	3.2.2, 3.3.2	**[CP]**
3.3.5	API Specification	Quân	4h	3d	2026-11-13	2026-11-16	3.3.4	**[CP]**

	for F06 Book Reservation							
3.4.1	UI Design System & Component Library	Ánh	5h	4d	2026-11-08	2026-11-12	M2 (PA2)	-
3.4.2	High-Fidelity Mockups for Key Workflows	Ánh	6h	4d	2026-11-12	2026-11-16	3.4.1	-
3.5.1	Design Review & Traceability Verification	Vinh	4h	3d	2026-11-15	2026-11-18	3.1.2, 3.2.2, 3.3.1, 3.3.4, 3.3.5, 3.4.2	**[CP]**
PA4	**Phase PA4 Timeline**	**Team**	**41h**	6d	2026-11-19	2026-11-25	**-**	**[CP]**
4.1.1	Master Test Plan Authoring	Vinh	5h	2d	2026-11-19	2026-11-21	M3 (PA3)	**[CP]**
4.1.2	Test Automation Framework Setup	Quân	4h	3d	2026-11-19	2026-11-22	M3 (PA3)	-
4.2.1	Test Cases for F01 Authenticatio n & Roles	Vinh	5h	2d	2026-11-20	2026-11-22	4.1.1	-
4.2.2	Test Cases for F02 Catalog &	Vinh	5h	2d	2026-11-20	2026-11-22	4.1.1	-

	F03 Copy Management							
4.2.3	Test Cases for F04 Search & Browse	Vinh	4h	2d	2026-11-21	2026-11-23	4.2.2	-
4.2.4	Test Cases for F05 Borrow & Return	Vinh	6h	2d	2026-11-21	2026-11-23	4.1.1, 4.2.2	**[CP]**
4.2.5	Test Cases for F06 Book Reservation	Vinh	5h	2d	2026-11-22	2026-11-24	4.2.4	**[CP]**
4.3.1	Test Traceability Matrix & Test Data Specification	Nguyên	4h	2d	2026-11-22	2026-11-24	4.2.1, 4.2.2, 4.2.3, 4.2.4, 4.2.5	**[CP]**
4.3.2	PA4 Document Review & Baseline	Vinh	3h	2d	2026-11-23	2026-11-25	4.1.2, 4.3.1	**[CP]**
PA5	**Phase PA5 Timeline**	**Team**	**91h**	13d	2026-11-26	2026-12-09	**-**	**[CP]**
5.1.1	Backend Boilerplate & DB Connection Setup	Quân	5h	2d	2026-11-26	2026-11-28	M3 (PA3), M4 (PA4)	**[CP]**
5.1.2	Frontend Boilerplate & Layout Setup	Quân	5h	2d	2026-11-26	2026-11-28	M3 (PA3)	-

5.2.1	F01 Authentication & Role-Based Access Control	Quân	6h	2d	2026-11-27	2026-11-29	5.1.1, 5.1.2	**[CP]**
5.2.2	F02 Book Catalog Management Module	Quân	6h	3d	2026-11-28	2026-12-01	5.2.1	**[CP]**
5.2.3	F03 Book Copy Management Module	Quân	5h	3d	2026-11-29	2026-12-02	5.2.2	**[CP]**
5.2.4	F04 Search & Browse Books Module	Ánh	6h	3d	2026-11-30	2026-12-03	5.2.2, 5.2.3	-
5.2.5	F05 Borrow & Return Management Module	Quân	8h	3d	2026-12-01	2026-12-04	5.2.3	**[CP]**
5.2.6	F06 Book Reservation Module	Quân	7h	3d	2026-12-02	2026-12-05	5.2.5	**[CP]**
5.2.7	Reader Dashboard & Active Loans View	Ánh	5h	2d	2026-12-03	2026-12-05	5.2.4, 5.2.5, 5.2.6	-
5.3.1	Automated Unit & Integration Testing	Quân	6h	3d	2026-12-03	2026-12-06	5.2.1, 5.2.2, 5.2.3,	**[CP]**

							5.2.5, 5.2.6	
5.3.2	Manual System Testing & Bug Logging	Vinh	7h	3d	2026-12-04	2026-12-07	5.2.7, 5.3.1	**[CP]**
5.3.3	Bug Fixing & Regression Verification	Quân	6h	3d	2026-12-05	2026-12-08	5.3.2	**[CP]**
5.3.4	Test Summary Report & QA Sign-off	Vinh	4h	2d	2026-12-07	2026-12-09	5.3.3	-
5.4.1	Cloud DB & Backend Deployment on Render	Quân	4h	2d	2026-12-06	2026-12-08	5.3.1, 5.3.3	**[CP]**
5.4.2	Frontend Cloud Deployment on Vercel	Quân	4h	2d	2026-12-06	2026-12-08	5.4.1	**[CP]**
5.4.3	Production Smoke Testing & Deployment Documentation	Vinh	4h	2d	2026-12-07	2026-12-09	5.4.2	**[CP]**
5.4.4	PA5 Submission & Code Baseline	Vinh	3h	1d	2026-12-08	2026-12-09	5.3.4, 5.4.3	**[CP]**

Final	Phase Final Timeline	Team	16h	6d	2026-12-10	2026-12-16	-	[CP]
6.1.1	Demo Scenario Scripting & Seed Data Preparation	Nguyên	4h	2d	2026-12-10	2026-12-12	M5 (PA5)	**[CP]**
6.1.2	Demo Rehearsal & Backup Recording	Vinh	3h	2d	2026-12-12	2026-12-14	6.1.1	**[CP]**
6.2.1	Presentation Slide Deck Creation	Ánh	5h	3d	2026-12-11	2026-12-14	M5 (PA5)	-
6.2.2	Final Contribution Assessment & Project Retrospective	Vinh	4h	2d	2026-12-14	2026-12-16	6.1.2, 6.2.1	**[CP]**

4.4.4 Critical Path Analysis & Schedule Risk Management

The Critical Path (CP) represents the sequence of dependent work packages that directly dictates the overall project completion timeline. A delay in any critical path package directly delays the project phase gate milestone. In UniLib, the critical path rigorously flows through:

- Phase 1 (Initiation): 1.1.1 (Kickoff) → 1.2.1 (Vision) → 1.2.2 (Scope) → 1.2.6 (WBS) → 1.2.8 (Schedule) → 1.3.3 (PA1 Baseline Review) → Milestone M1 (Oct 21).
- Phase 2 (Requirements): 2.1.1 (Domain Glossary) → 2.1.2 (Use Case Model) → 2.2.5 (F05 Borrow/Return Spec) → 2.2.6 (F06 Reservation Spec) → 2.4.1 (RTM Baseline) → 2.4.2 (PA2 Review) → Milestone M2 (Nov 04).

· Phase 3 (Architecture & Design): 3.1.1 (C4 Architecture) → 3.2.1 (ERD) → 3.2.2 (Physical DDL Schema) → 3.3.4 (F05 Circulation API) → 3.3.5 (F06 Reservation API) → 3.5.1 (PA3 Review) → Milestone M3 (Nov 18).

· Phase 4 (Test Planning): 4.1.1 (Master Test Plan) → 4.2.4 (F05 Circulation Test Cases) → 4.2.5 (F06 Reservation Test Cases) → 4.3.1 (Test Traceability) → 4.3.2 (PA4 Review) → Milestone M4 (Nov 25).

· Phase 5 (Implementation & Deployment): 5.1.1 (Backend & DB Setup) → 5.2.1 (F01 Auth & RBAC) → 5.2.2 (F02 Catalog) → 5.2.3 (F03 Copy Inventory) → 5.2.5 (F05 Borrow & Return Module) → 5.2.6 (F06 Reservation Module) → 5.3.1 (Integration Testing) → 5.3.2 (System Testing) → 5.3.3 (Bug Fixing) → 5.4.1 (Render DB Deploy) → 5.4.2 (Vercel Deploy) → 5.4.3 (Smoke Testing) → 5.4.4 (PA5 Submission) → Milestone M5 (Dec 09).

· Phase 6 (Final Demo & Closeout): 6.1.1 (Demo Script & Seed Data) → 6.1.2 (Demo Rehearsal) → 6.2.2 (Final Retrospective & Submission) → Milestone M6 (Dec 16).

Float / Slack Analysis: Non-critical paths possess flexible schedule float to absorb variances:

· Hardware & Environment (1.2.4): 2 days float.

· Wireframes (2.3.1, 2.3.2): 2 days float before PA2 review.

· UI Design System & Mockups (3.4.1, 3.4.2): 2 days float before design baseline.

· Reader Dashboard (5.2.7): 3 days float before system test freeze.

· Presentation Slides (6.2.1): 2 days float before final delivery.

Total Critical Path Duration: 63 calendar days across 9 weeks (from 2026-10-15 to 2026-12-16).

4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	Nguyễn Phan Thế Vinh	Trần Hà Anh Quân	Overall project planning, schedule monitoring, task tracking on GitHub Projects, risk management, internal team meetings, weekly status reports, primary communication point with TA/Lecturer.
Business analyst	Nguyễn Thanh Thảo Nguyên	Vũ Thị Ánh	Eliciting and documenting functional/non-functional requirements (SRS), defining domain entities, use case modeling, business rules specification (Loan, Reservation, Overdue), maintaining requirements traceability matrix.

Designer / Architect	Vũ Thị Ánh	Nguyễn Thanh Thảo Nguyên	System architecture design (C4 diagrams), UI/UX wireframes and high-fidelity mockups, relational database schema modeling (PostgreSQL ERD), design consistency, and component styling.
Developers	Trần Hà Anh Quân	Nguyễn Phan Thế Vinh	Frontend (React) and backend (Node.js/Express) implementation, database schema migrations, REST API development, unit testing, CI/CD pipeline, cloud deployment (Vercel/Render), and code review.
Tester	Nguyễn Phan Thế Vinh	Nguyễn Thanh Thảo Nguyên	Master Test Plan development, test case design for all Must-have features (F01–F06), test execution (functional, integration, system), defect tracking on GitHub Issues, and test summary reporting.

4.6 Human Resources and Cost Estimate

4.6.1 Human Resources & Labor Cost Model

The project human resource cost is computed based on bottom-up estimated effort and a realistic junior software engineer / QA tester market hourly rate in Vietnam. A standard benchmark rate of 100,000 VND/hour (equivalent to approximately $4.00 USD/hour at an exchange rate of 1 USD = 25,000 VND) is applied across all team members.

Team Member	Primary Role(s)	Effort (h)	Hourly Rate	Labor Cost (VND)	Labor Cost (USD)
Nguyễn Phan Thế Vinh	Team Lead / PM, Tester	96	100,000 VND/h	9,600,000 VND	$384.00
Nguyễn Thanh Thảo Nguyên	Business Analyst	54	100,000 VND/h	5,400,000 VND	$216.00
Vũ Thị Ánh	Designer / Architect	64	100,000 VND/h	6,400,000 VND	$256.00
Trần Hà Anh Quân	Developer	104	100,000 VND/h	10,400,000 VND	$416.00

Total Labor Cost	**Entire Team (4 Members)**	**318**	**100,000 VND/h**	**31,800,000 VND**	**$1,272.00**

4.6.2 Tools, Hosting & Infrastructure Cost

The project prioritizes free and open-source tooling for development and CI/CD. However, to guarantee reliable live demonstration without spin-down latency and to guard against Render's 30-day PostgreSQL database expiration, a dedicated hosting buffer is provisioned for Weeks 9–11 (approx. 1 month).

Tool / Service	Tier / Plan	Purpose / Usage	Cost (VND)	Cost (USD)
GitHub	Free Tier	Private repository, GitHub Projects task board, GitHub Actions (2,000 free CI minutes/mo)	0 VND	$0.00
Vercel	Hobby (Free)	React frontend continuous deployment, global CDN, automated PR preview deployments	0 VND	$0.00
Render (Development)	Free Tier	Express backend API web service and PostgreSQL testing instance (W1–W8)	0 VND	$0.00
Render Starter Provisioning	Paid Starter ($7/mo + $7/mo)	Production deployment buffer for Weeks 9–11: dedicated instance to avoid 50s cold-start spin-down and prevent 30-day database deletion during demo grading	350,000 VND	$14.00
Total Tooling & Hosting	**Combined Infrastructure**	**Complete development, CI/CD, and production deployment infrastructure**	**350,000 VND**	**$14.00**

4.6.3 Total Project Cost Summary & Contingency Reserve

To mitigate technical and estimation risks (such as unexpected bug fixing or integration rework during PA5), a 10% Contingency Reserve is incorporated into the overall financial model.

Cost Category	Description	Cost (VND)	Cost (USD)	% of Total
Direct Labor Cost	318 person-hours @ 100,000 VND/hour across 4 team members	31,800,000 VND	$1,272.00	89.9%
Tooling & Hosting	GitHub, Vercel, and Render production demo instance buffer	350,000 VND	$14.00	1.0%
Contingency Reserve	10% management reserve for defect fixing and integration buffers	3,215,000 VND	$128.60	9.1%
Total Estimated Budget	**Comprehensive UniLib project investment baseline**	**35,365,000 VND**	**$1,414.60**	**100.0%**

5 Risk Management

Risk Measurement Scale

In accordance with standard Software Engineering and Project Management methodologies (Sommerville / Pressman), risks are evaluated quantitatively across two dimensions: Probability (P) and Loss (L), yielding Risk Exposure (RE = P × L):

Probability (P): 1 – Very Low (< 10%), 2 – Low (10%–25%), 3 – Moderate (26%–50%), 4 – High (51%–75%), 5 – Very High (> 75%).

Loss / Impact (L): 1 – Insignificant (< 2h rework), 2 – Minor (absorbable within float), 3 – Moderate (noticeable impact, requires management attention), 4 – Major / Critical (jeopardizes milestone exit criteria), 5 – Catastrophic (project failure, unrecoverable breach).

Risk Exposure (RE = P × L, scale 1–25): 1–5 (Low Risk, monitored periodically during weekly syncs), 6–12 (Moderate to High Risk, proactive mitigation required), 13–19 (High Risk, critical threat requiring primary owner oversight), 20–25 (Critical / Extreme Risk, structural replanning required).

Master Risk Register Table

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R1	Data Inconsistency in Copy, Loan, and Reservation State Transitions (Concurrent circulation events cause race conditions, corrupting state synchronization across BookCopy, Loan, and Reservation entities)	Technology	3 (Moderate, ~35%)	4 (Critical)	12 (High)	Enforce strict atomic database transactions (BEGIN ... COMMIT) with row-level locking (SELECT ... FOR UPDATE) in PostgreSQL. Formally model FSM for Copy, Loan, and Reservation lifecycles in PA2/PA3. Implement automated concurrency integration test suites (Jest +	Quân (Dev) / Ánh (Architect)

						Supertest) in PA4/PA5 before merge.	
R2	Cloud Deployment Latency, Platform Cold-Starts, and Ephemeral DB Expiration (Render free web service 50s cold-start delay and 30-day PostgreSQL database expiration risk demo timeouts or test data loss during grading)	Technology	4 (High, -60%)	3 (Moderate)	12 (High)	Deploy deployment pipeline early in PA5 (WBS 5.4.1/5.4.2). Provision dedicated Render Starter buffer (budget in Section 4.6.2) for Weeks 9–11. Maintain idempotent database migration and seed scripts (npm run db:seed) to recreate schema and demo data in < 2 min. Prepare backup video demo (WBS 6.1.2).	Quân (Dev)
R3	Key Team Member Unavailability or Illness During Peak Implementation (PA5) (PA5 implementation requires 91 person-hours within 2 weeks. Illness or exam conflicts for key contributors (Quân as lead dev or Vinh as tester/PM) would block critical path)	People	2 (Low, -20%)	4 (Critical)	8 (Moderate)	Strictly enforce Primary–Backup Role Matrix (Section 4.5: Vinh backs up Quân; Ánh backs up Nguyên; Quân backs up Vinh). Maintain explicit ADRs, clean modular code, and conventional PR descriptions. Hold daily 5-min async check-ins during PA5; reallocate non-critical UI dashboard tasks to backup members.	Vinh (PM / Tester)
R4	Scope Creep Beyond PA0 Baseline & Out-of-Scope Feature Requests (Attempting to introduce features outside approved PA0 scope: e-book reading, online payments, hardware barcode/RFID scanners, external email/SMS APIs, mobile app)	Requirements	3 (Moderate, -30%)	4 (Critical)	12 (High)	Strictly enforce MoSCoW scope baseline from Section 2.4 (F01–F06 Must-haves only). Maintain strict RTM starting in PA2 (WBS 2.4.1). Require formal Change Request review with unanimous team approval. Restrict reservation notices to in-app	Nguyên (BA)

						status indicators without external email/SMS.	
R5	Underestimation of Circulation Workflow Complexity & Integration Effort (Implementation effort for interconnected library circulation workflows (F05 loan checkout/return, overdue tracking, and F06 FIFO reservations) exceeds estimated hours)	Estimation	3 (Moderate, -40%)	3 (Moderate)	9 (Moderate)	Applied PERT Three-Point estimation with pessimistic variance (P = 10h on F05) and incorporated 10% contingency reserve (.60 / 3,215,000 VND). Implement core backend APIs and unit tests first before UI. Utilize built-in schedule float (2–3 days); descope optional UI animations if needed.	Vinh (PM / Tester)
R6	University Midterm Examination & Multi-Course Assignment Deadline Collisions (Academic calendar clash during Weeks 6–8 with midterms and competing coursework across other subjects, causing potential team bandwidth reduction and missed internal deadlines)	Organisational	4 (High, -65%)	3 (Moderate)	12 (High)	Front-load deliverables: complete requirements (PA2) and architecture/DDL schemas (PA3) early in Weeks 3–5 before exam peaks. Enforce a 2-day internal deadline rule. Review member exam schedules during Thursday 20:30 weekly syncs and adjust work package pacing on GitHub Projects accordingly.	Vinh (Team Lead / PM)

6 Tools & Collaboration Setup

Tools:

- ✔ Moodle – announcements and PA submissions.
- ✔ Communication – Zalo
- ✔ Task board – GitHub Projects
- ✔ Git repository – GitHub (recommended), GitLab or Bitbucket; the instructor and TA have access:
  - main branch is protected; changes are merged through pull requests with at least one review
  - README (description, setup), .gitignore; never commit secrets (.env, API keys, passwords)
  - commit messages follow the Conventional Commits style (feat:, fix:, docs:, test:, ...)
- ✔ Weekly status meeting (15 minutes) and weekly status report (template provided), committed to /docs/management/weekly every week from week 2 to week 10. Weekly deadlines: 23:59, Friday

Repository structure:

Folder	Content
/src	Source code
/docs/management	Project plan; weekly status reports in /docs/management/weekly
/docs/requirements	Vision, SRS, use cases, mockups
/docs/design	Architecture (C4), ADRs, UML models, UI design
/docs/test	Test plan, test cases, test reports
/docs/ai-usage	AI usage logs

/pa/PA1 ... /pa/PA5	One sub-folder per PA submission (PDF + source files)

7 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Phan Thế Vinh	Google DeepMind Agent (Antigravity AI)	Structured the 3-level WBS, drafted phase gate exit criteria, and formulated kickoff meeting minutes (ACT-01, ACT-04 / WBS 1.1.1, 1.2.5, 1.2.6).	Verified against PA0 Must-have features (F01–F06), aligned effort hours (2–8h per package), and ensured role assignments strictly reflect the Team Contract. Reconciled package decomposition to eliminate overlapping boundaries.
2	Trần Hà Anh Quân	OpenAI Codex	Initialized repository skeleton (/src, /docs, /pa), .gitignore, PR template, CI workflow (ci.yml), Dependabot, Copilot instructions, and tooling documentation (ACT-08 / WBS 1.3.1).	Automated local checks verified all 11 required template directories, 15 ignore/keep rules, relative doc links, and 3 CI shell checks. Flagged administrator requirements for branch protection and TA access without committing credentials.
3	Trần Hà Anh Quân	OpenAI Codex	Drafted Section 3.2 Technology Stack proposal and created 69 GitHub Issues (#6–#74) from WBS dictionary for task board setup (ACT-08 / WBS 1.2.3, 1.3.2).	Compared stack proposal with PA0 constraints; consulted official tool/hosting documentation (Vite, Express, PostgreSQL, Render); verified 69 issues against WBS; flagged 318h effort discrepancy against 274h placeholder for PM resolution.
4	Nguyễn Phan Thế Vinh	Google DeepMind Agent (Antigravity AI)	Formulated PERT three-point estimates, scheduled 69 WBS packages across 11 weeks, identified critical path, and calculated human resource and tooling cost model (ACT-05, ACT-06 / WBS 1.2.7, 1.2.8).	Reconciled 318 person-hours across team capacity (7.23 h/person/week within 6–8h guideline), verified milestone dates against course calendar (PA1–PA5 and Final demo), ensured critical path logic aligns with architecture dependencies, and verified hosting cost against Render/Vercel pricing models. Technical estimates reviewed by Developer (Quân).
5	Vũ Thị Ánh	ChatGPT, Gemini	Assisted in authoring preliminary C4 System Context architecture (Section 3.1), specifying hardware and operating environment constraints (Section 3.3), and	Cross-checked task dependencies, phase durations, and critical path against course timeline and team bandwidth. Validated architecture diagrams and browser compatibility (HTML5 evergreen browsers, no

			constructing the schedule network diagram and Gantt chart (ACT-03, ACT-06 / WBS 1.2.3, 1.2.4, 1.2.8).	mobile apps) with PM and Developer to ensure full compliance with PA0 constraints.
6	Nguyễn Thanh Thảo Nguyên	ChatGPT, Gemini	Assisted in structuring the preliminary problem statement (Section 2.1), product positioning statement (Section 2.2), stakeholder analysis (Section 2.3), and MoSCoW scope classification (ACT-02 / WBS 1.2.1, 1.2.2).	Reviewed and edited all content from a Business Analyst perspective; strictly enforced out-of-scope boundaries (no RFID, no online payment, no DRM); verified that all 6 Must-have features align with PA0; eliminated ambiguities in loan/reservation rules and flagged open items for team confirmation.
7	Nguyễn Phan Thế Vinh	Google DeepMind Agent (Antigravity AI)	Formulated Section 5 Risk Management with quantitative RE = P x L and mitigations, consolidated master AI Usage Log across all 4 members, updated Section 1 Member Contribution with evidence links, and executed Phase Gate M1 GO/NO-GO review (ACT-01, ACT-07, ACT-09 / WBS 1.2.9, 1.3.3).	Verified risk exposure calculations using standard 1–5 probability and impact scales; ensured domain-specific risks (Copy/Loan/Reservation states, cloud deployment/expiration) are thoroughly addressed; cross-referenced all member AI logs with repository artifacts and commits; confirmed that contribution percentages total exactly 100% (25% each) with unanimous consensus.